Graph Neural Networks for Molecular Structure Learning

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- Molecules represented as Graphs
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  - Molecule Classification task
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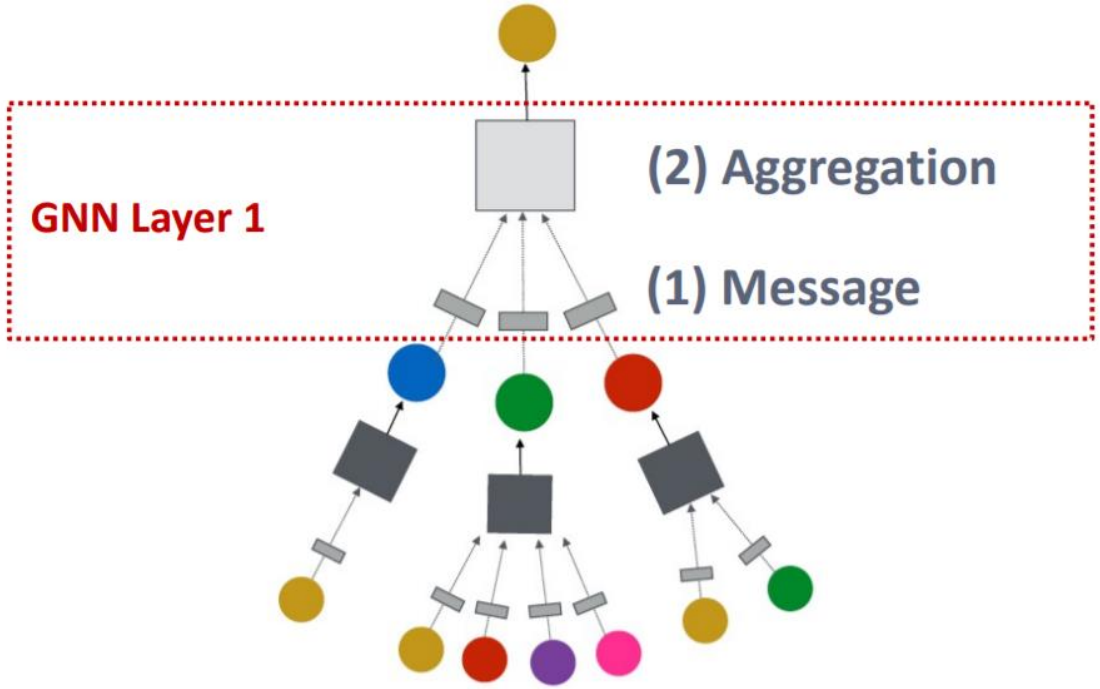
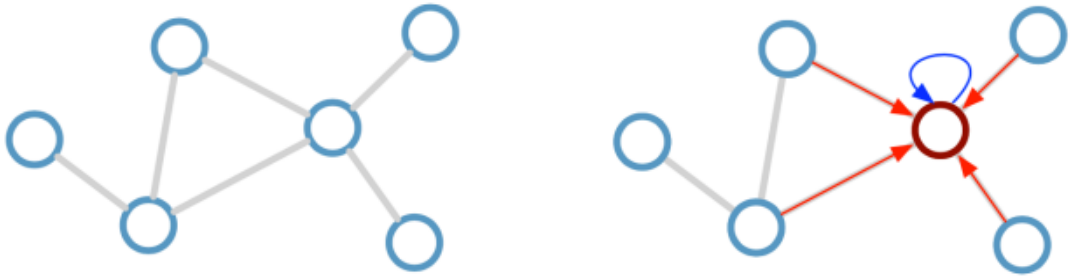
- Recent Advances in GNNs on Molecular Learning

- Practice code
  - Read molecules from SMILE strings, and visualize as 2D and 3D graphs
  - Apply GNNs models (GCN, GraphSAGE, GIN, GAT, GT) on benchmark molecular datasets on classification and regression tasks; and performance comparison
  - Benckmarks: MoleculeNet

Recap: Graph Convolutional Networks (GCNs)

- GNN Layer = **Message + Aggregation**
  - Message COMPUTATION
    - How to make each neighbourhood node as embedding?
  - Message AGGERGATION
    - How to combine those embeddings?





Update rule: $$\mathbf{h}_i^{(l+1)} = \sigma\left(\mathbf{h}_i^{(l)}\mathbf{W}_0^{(l)} + \sum_{j\in\mathcal{N}_i}\frac{1}{c_{ij}}\mathbf{h}_j^{(l)}\mathbf{W}_1^{(l)}\right)$$

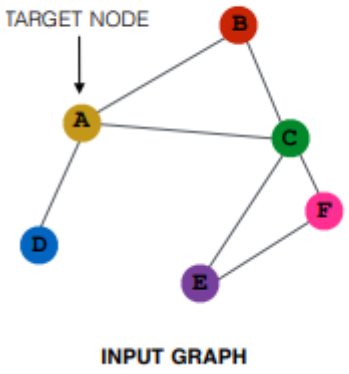
Kipf & Welling (ICLR 2017), related previous works by Duvenaud et al. (NIPS 2015) and Li et al. (ICLR 2016)

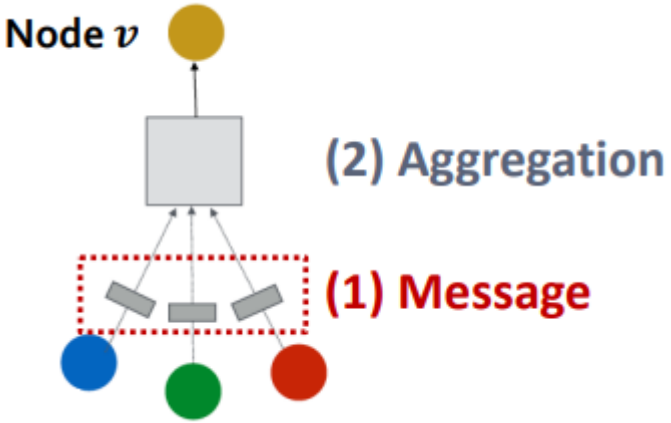
GNN Layer: Message Computation

- Intuition: Each node will create a message, which will be sent to other nodes later

- Example: A Linear layer $\mathbf{m}_u^{(l)} = \mathbf{W}^{(l)}\mathbf{h}_u^{(l-1)}$
  - Multiply node features with weight matrix $\mathbf{W}^{(l)}$

Message function: $\mathbf{m}_u^{(l)} = \mathrm{MSG}^{(l)}\left(\mathbf{h}_u^{(l-1)}\right)$





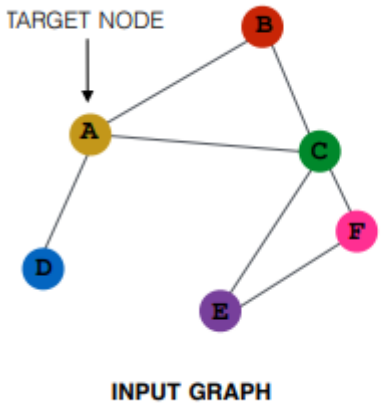
GNN Layer: Message Aggregation

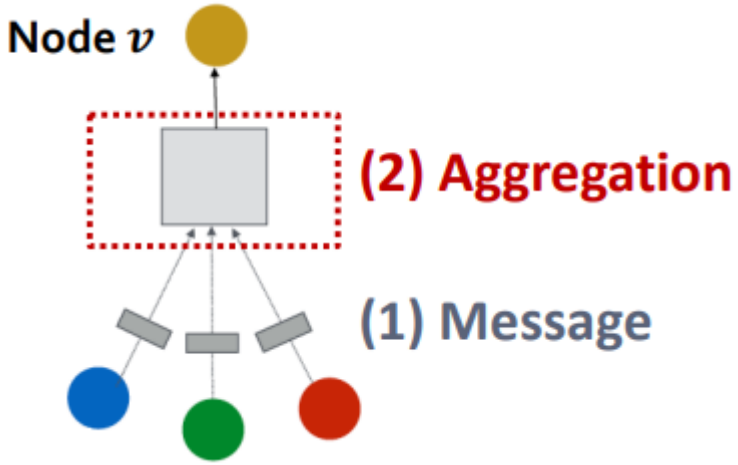
- Intuition: Each node will aggregate the messages from node v's neighbors

$$\mathbf{h}_v^{(l)} = \mathrm{AGG}^{(l)}\left(\left\{\mathbf{m}_u^{(l)}, u \in N(v)\right\}\right)$$

- Example: Sum(·), Mean(·) or Max(·) aggregator

$$\mathbf{h}_v^{(l)} = \mathrm{Sum}(\{\mathbf{m}_u^{(l)}, u \in N(v)\})$$

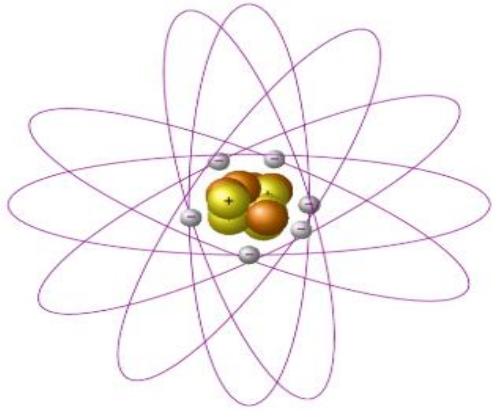




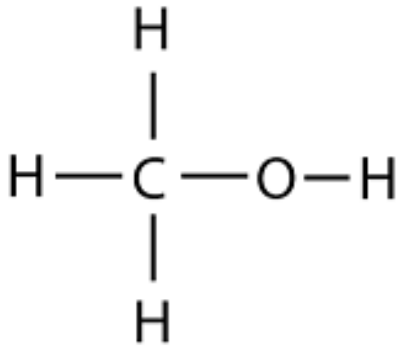
Introduction to Molecules

- Atoms are the smallest stable units of matter
- They can combine to form molecules with complex shapes
- The atomic components and unique structural shape of a particular molecule determine its chemical functions

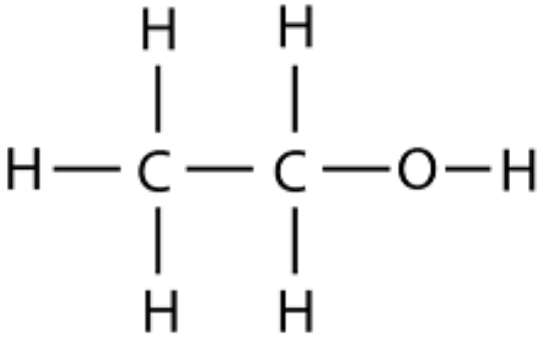
Carbon atom



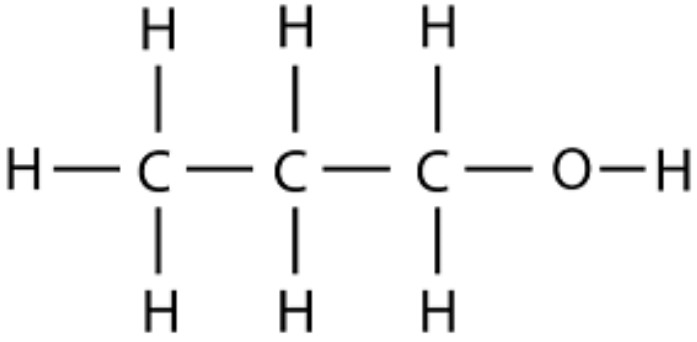
Methanol



Ethanol



Propanol

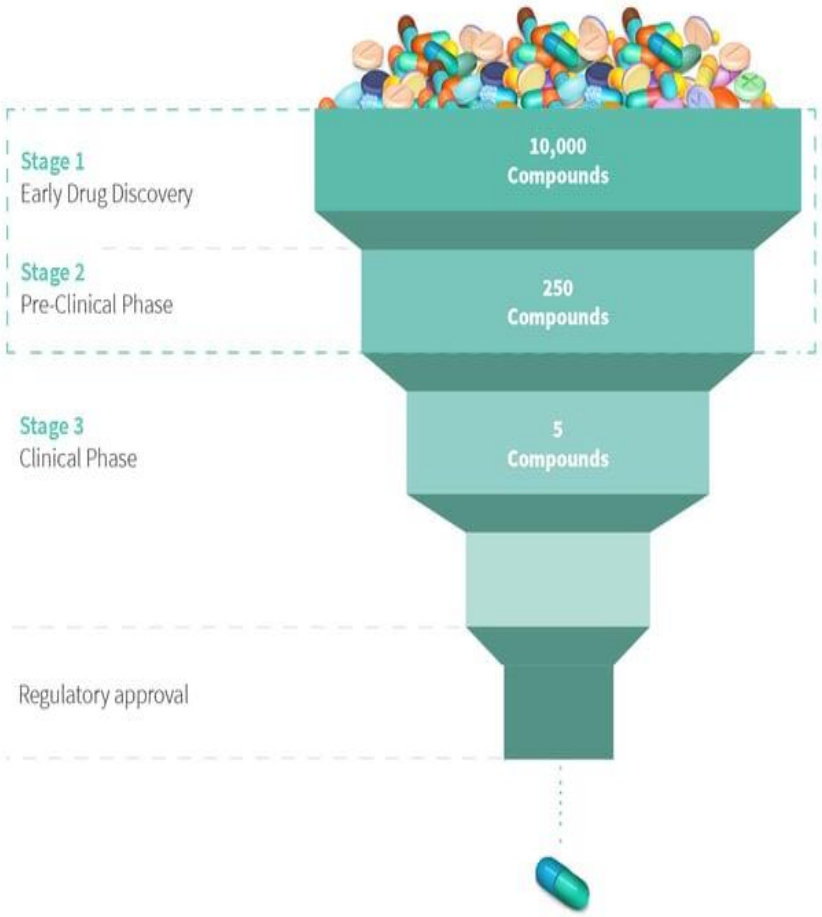


The alcohol group (-**OH group**) is made up of oxygen and hydrogen atoms.

Molecular Biology Applications

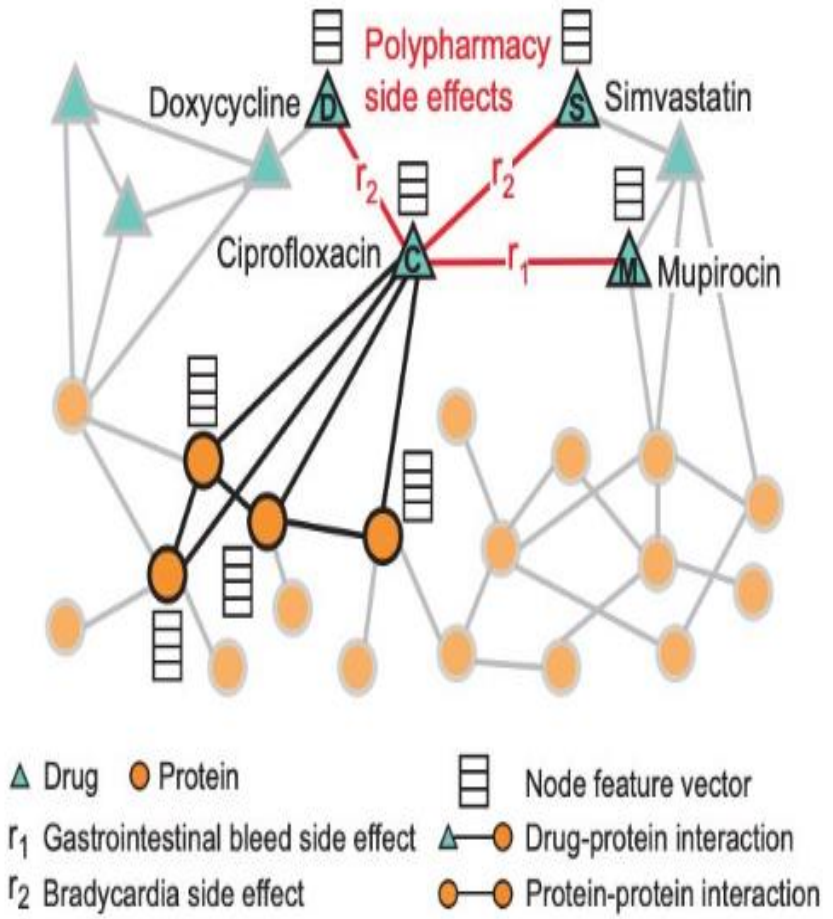
- Application: drug design is very important:
  - Drug discovery is costly, time-expensive
- On average, for every 10,000 structures synthesized during design:
  - 250 will reach animal testing
  - 5 will reach phase I clinical trials
  - Only 1 reach the marketplace

- **Our target:**
- Use molecular databases and ML-based models to support searching the complex space of candidate.
- Want targeted molecules that optimise for specific properties.



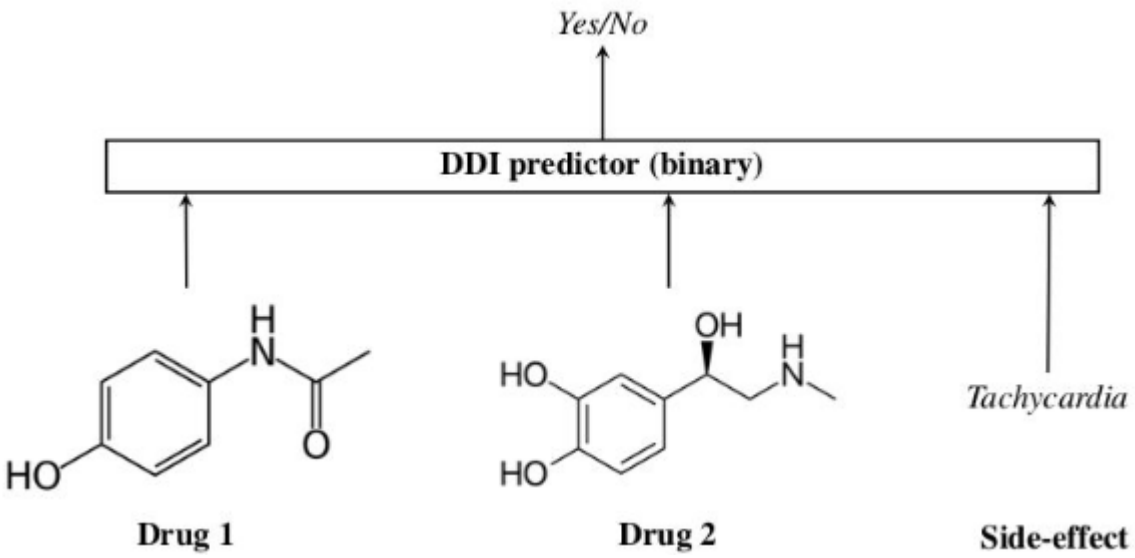
Molecular Biology Applications

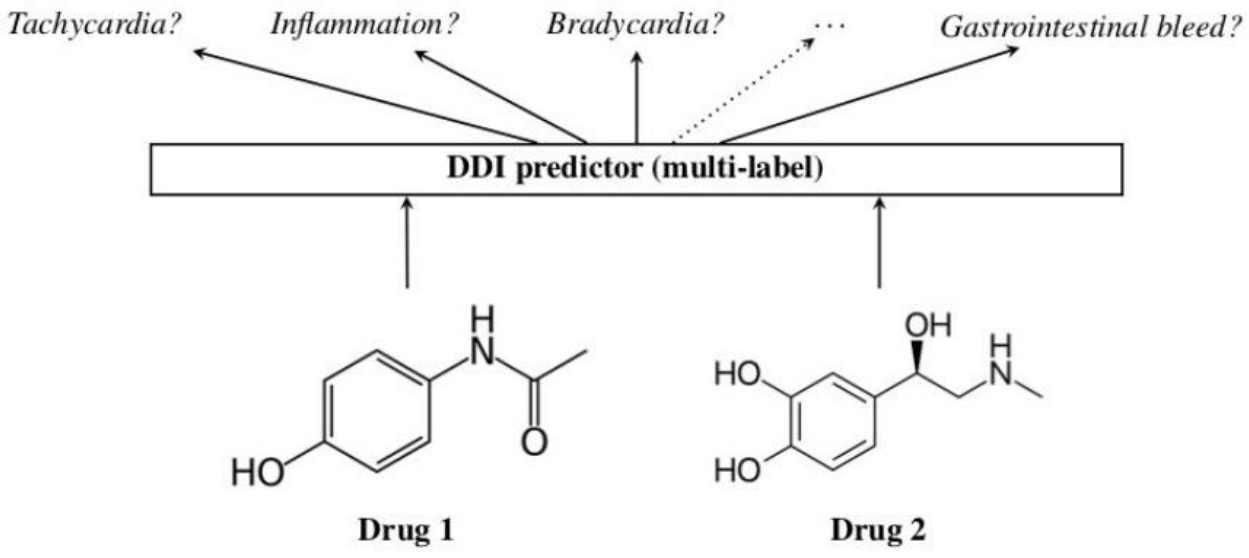
- The process of identifying interactions between drug combinations that may cause adverse side effects.
- Identifying how different drugs interact with each other for Patient safety & Treatment efficacy.
- **Our target:** Can model the complex associations between atoms and functional groups within and between drug molecules.



Molecular Biology Applications

- Drug-drug interaction (DDI) prediction
- We aim to identify unwanted side effects that can occur when multiple drugs are taken together
- For example:
  - Input: set of molecules w/wo side-effect
  - Output: predict the interaction (binary or multi-label classification)



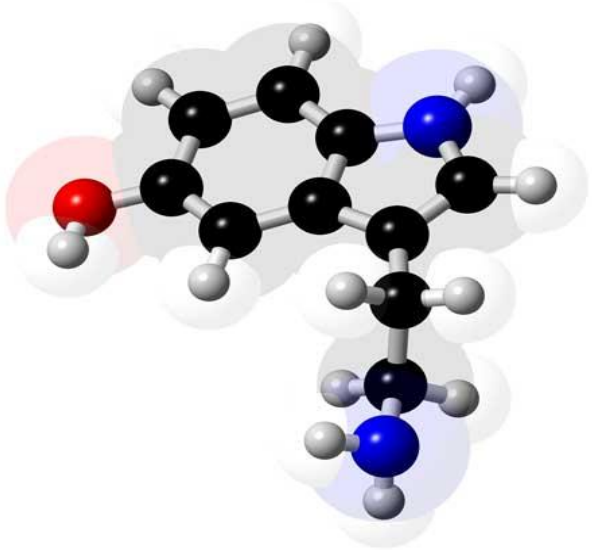


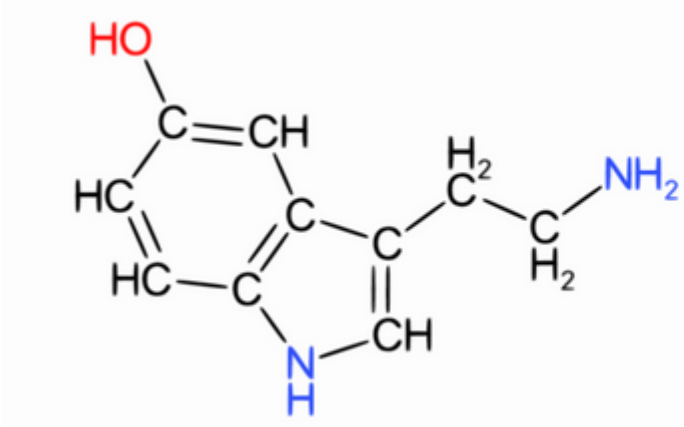
Representing molecules as graphs

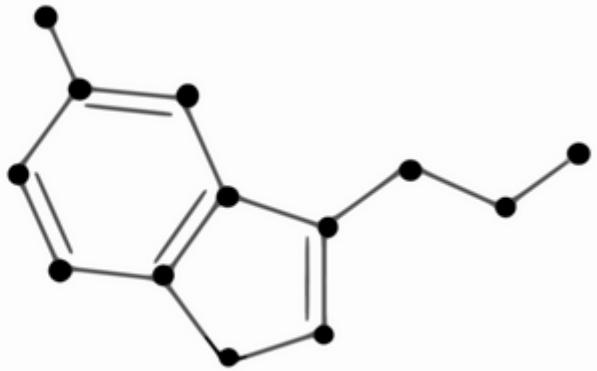
- A molecular graph is a labelled graph:
  - The nodes and edges represent the atoms and bonds of the molecule, respectively.

- Molecule graph provides direct access to the graph underpinning all molecule objects (atoms, functional groups), allowing seamless integration with existing graph functionality

  - E.g., Graph representation of the Serotonin ($C_{10}H_{12}N_2O$) molecule with functional groups, i.e., NH_2, NH, and $-OH$.

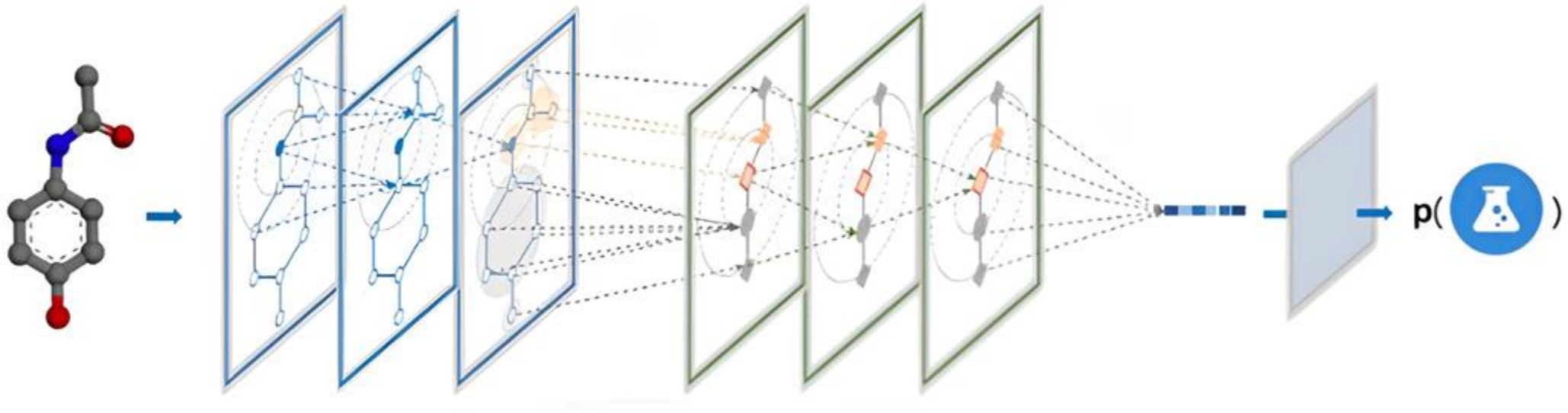






Molecular property prediction with GNNs: steps by steps

- Inputs:
  - Given a set of graph pairs $D = \{(G_1, Y_1), (G_2, Y_2), \cdots, (G_N, Y_N)\}$
- The goal is to train a GNN model M that predicts the target values of given arbitrary graph pairs in an end-to-end manner, i.e., $Y_i = M(G_i)$. The target Y is a scalar value for regression tasks, while it is a binary class label $Y \in \{0,1\}$ for (binary) classification tasks.

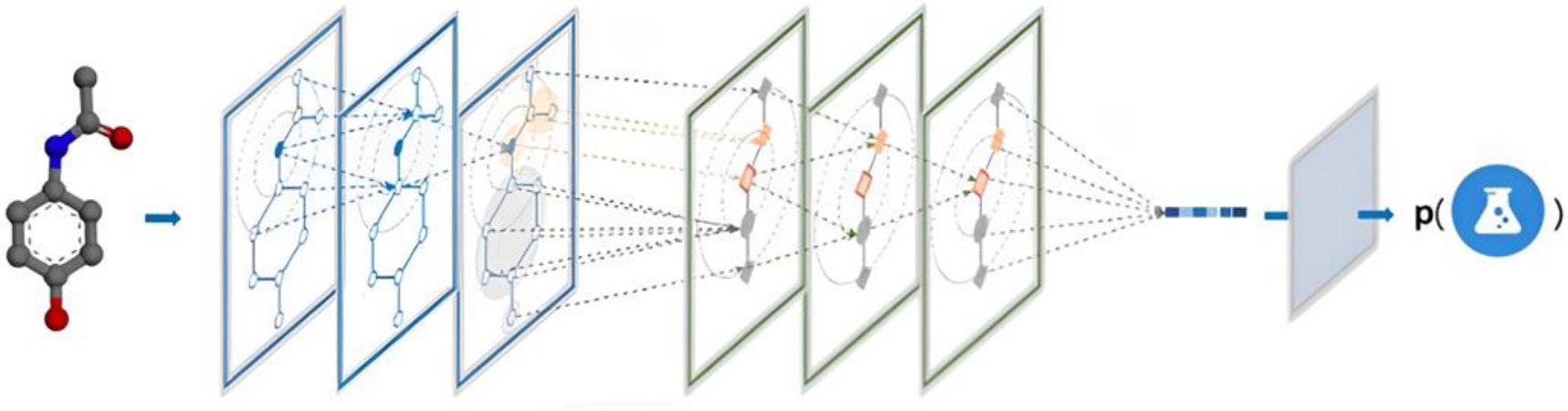


Molecules | Graph Neural Networks | Pooling Function | Task Prediction

Sanchez-Lengeling et al., Machine Learning for Scent: Learning Generalizable Perceptual Representations of Small Molecules (2019)

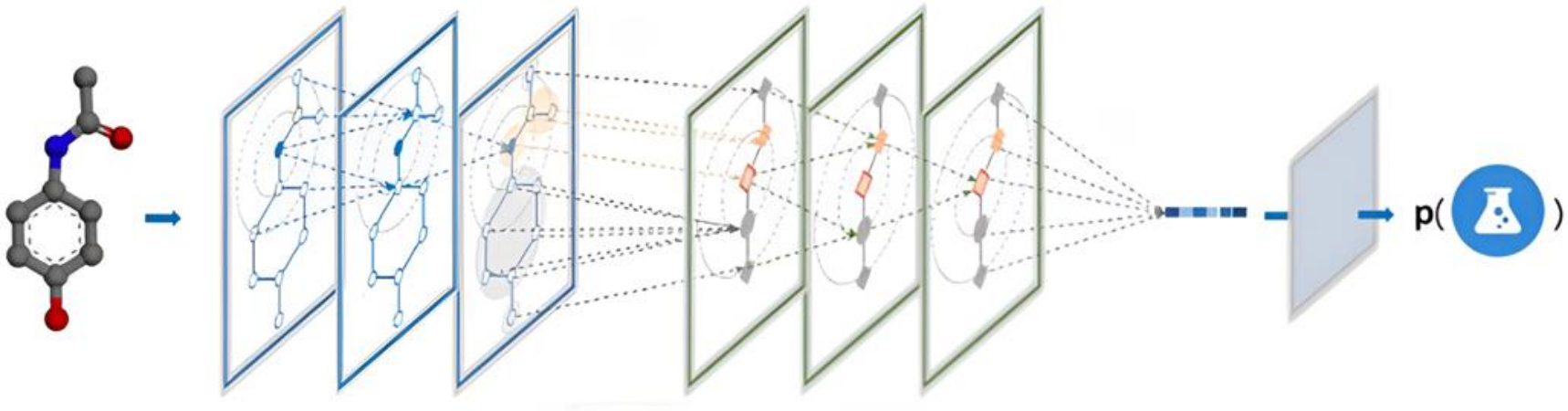
Molecular property prediction with GNNs: steps by steps

- Graph Representation of Molecules
  - **Nodes (Atoms)**: Each atom is represented as a node with features like atom type (e.g., carbon, oxygen), formal charge, hybridization, etc.
  - **Edges (Bonds)**: Bonds between atoms (e.g., single, double, aromatic) form the edges and carry properties that indicate bond strength, type, or length.
  - **Graph Construction**: The molecule's 2D or 3D structure is converted into a graph that GNNs can process, encoding molecular connectivity and interactions.



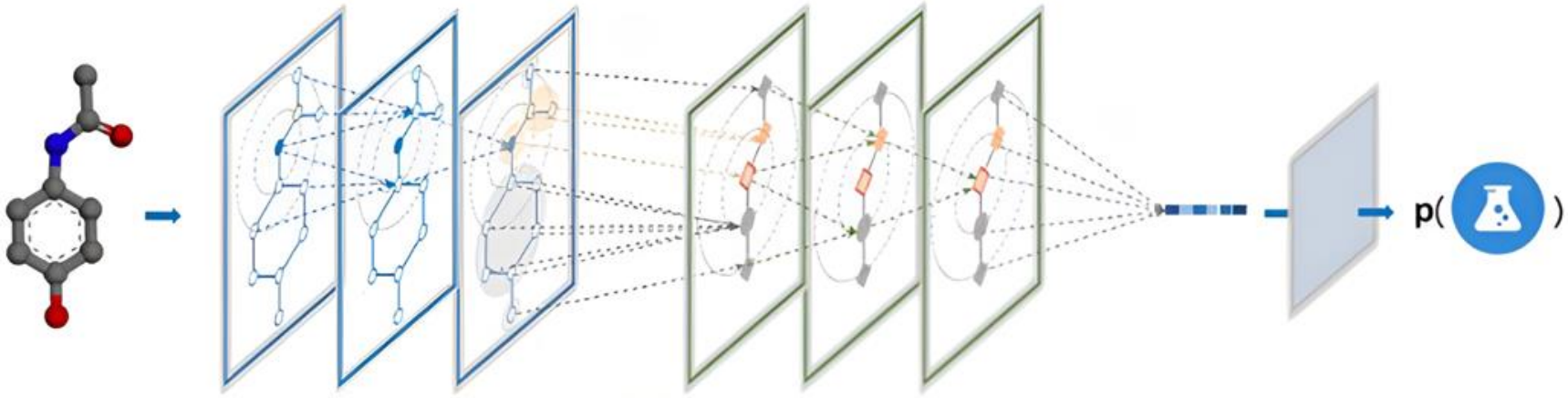
Molecular property prediction with GNNs: steps by steps

- Feature Extraction and Embedding Initialization
  - **Node Features**: Each atom's characteristics are embedded as vectors, capturing information such as atomic number, charge, and valency
  - **Edge Features**: Bond characteristics can also be embedded to inform GNNs about bond types, order, or lengths (if using 3D data)



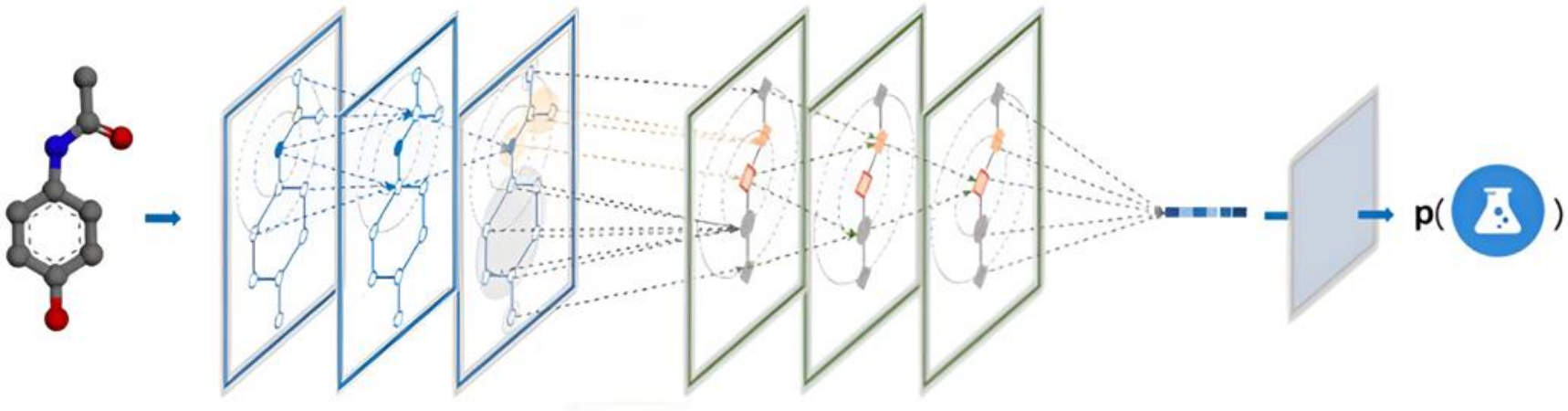
Molecular property prediction with GNNs: steps by steps

- Message Passing and Aggregation:
  - **Message Passing**: In each layer, nodes aggregate features from their neighbouring nodes, “passing messages” to capture local interactions and structural information
  - **Aggregation Function**: Common aggregation functions (mean, sum, max) help summarize neighbouring information, creating a new node embedding that reflects both the node’s features and its neighbourhood
  - **Multiple Layers**: Stacking multiple GNN layers allows the model to capture information from nodes further away, which helps in learning both local and global molecular structures



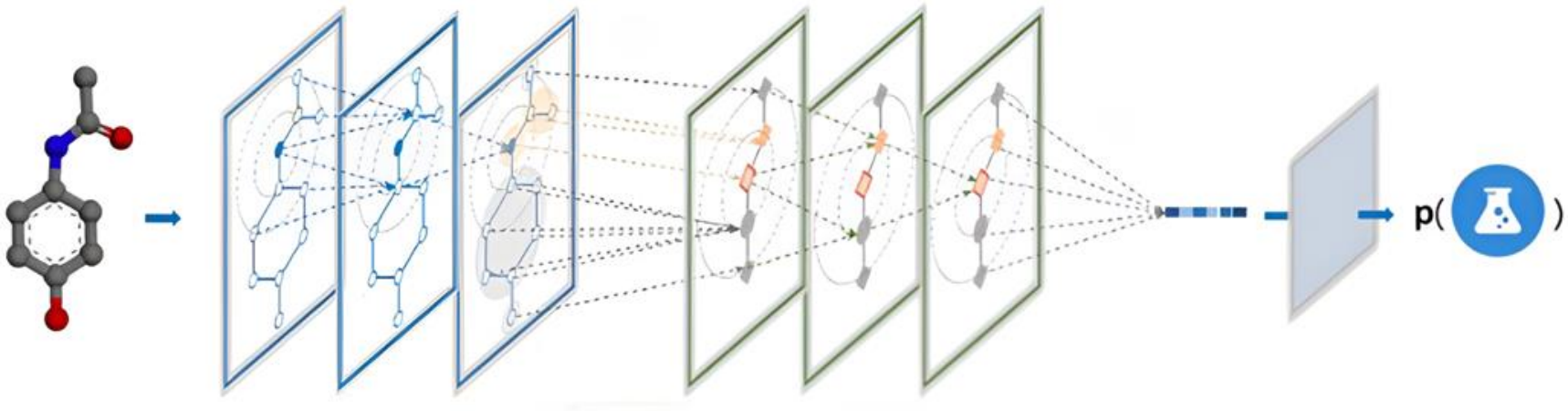
Molecular property prediction with GNNs: steps by steps

- Pooling and Readout:
  - **Global Pooling**: Since the final output is a single prediction per molecule, GNNs often use pooling methods to combine all node embeddings into a single graph-level representation. Pooling techniques like **mean pooling**, **max pooling**, or **attention pooling** summarize node embeddings into a fixed-size vector
  - **Hierarchical Pooling**: In complex molecules, hierarchical pooling (based on substructures or motifs) can capture higher-level chemical motifs, allowing the model to distinguish functional groups or rings



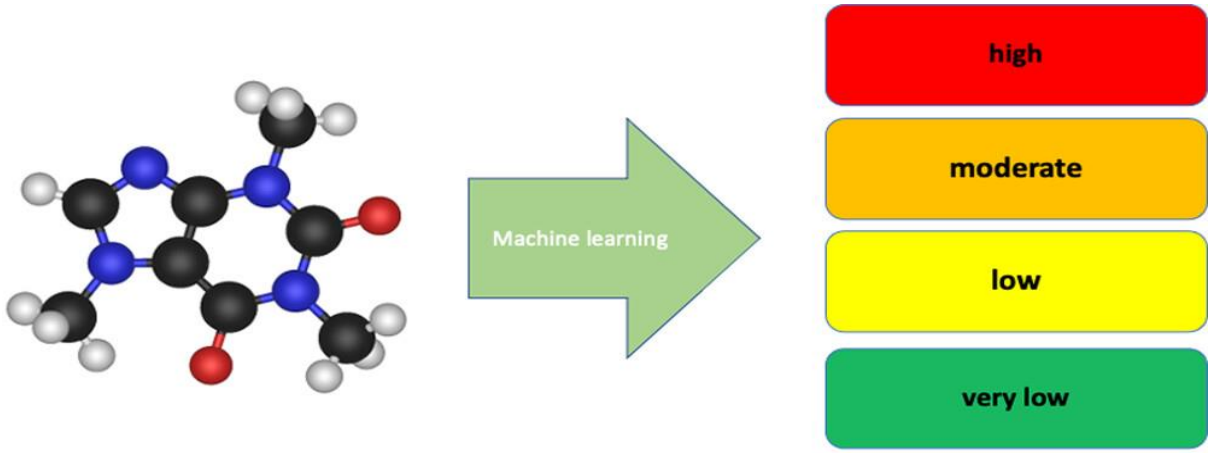
Molecular property prediction with GNNs: steps by steps

- Classification Layer:
  - **Fully Connected Layer**: The pooled embedding is typically passed through a fully connected layer for classification
  - **Output**: The model predicts a class label, such as “toxic” vs. “non-toxic” or the presence of a certain property (e.g., drug-like properties, fluorescence)



Molecule Classification Task

- **Goal**: Assign a categorical label to a molecule (e.g., toxic vs. non-toxic, active vs. inactive).
- **Applications**:
  - **Drug Discovery**: Predicting if a molecule is likely to be effective against a target disease
  - **Toxicology**: Classifying compounds as toxic or non-toxic to ensure safety in drugs or materials
  - **Materials Science**: Identifying molecules that meet specific properties like biodegradability, fluorescence, or electrical conductivity



Molecular descriptors

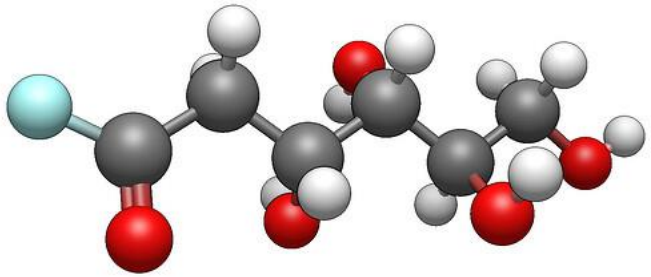
Acute fish toxicity categories

Molecule Property Prediction Task (Regression)

- **Goal**: Predict continuous properties of molecules, such as boiling point, binding affinity, or toxicity level
- Applications
  - **Chemical and Physical Property Prediction**: Estimating properties like melting point, solubility, or partition coefficient
  - **Drug Development**: Predicting binding affinities to determine how strongly a drug molecule interacts with a target

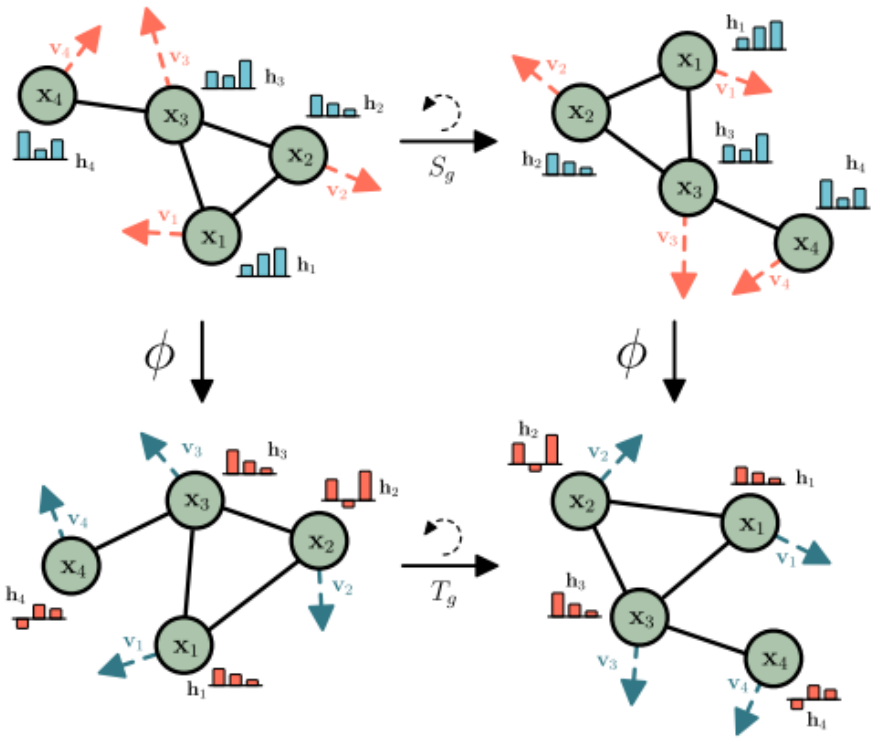
Equivarient GNNs

- Equivariant GNNs are designed to preserve certain symmetry properties, particularly rotation and translation, in their architecture.
- This is especially important in tasks involving 3D spatial data, such as molecular modelling or 3D object recognition, where the relationships between nodes must remain consistent despite changes in their spatial configurations.
- **Why 3D coordinates?**
  - The use of 3D coordinates is crucial for various tasks in molecular modelling and related fields, particularly when properties and behaviours of atoms and molecules are directly linked to their spatial arrangements.
  - In molecular learning, the spatial arrangement of atoms greatly influences their interactions, stability, and reactivity. Properties such as bond lengths, angles, and dihedral angles are inherently 3D.



Equivarient Graph Neural Networks

- Rotation equivariance in GNNs refers to the property that a model's output remains consistent under rotations of the graph structure
- Example of rotation equivariance on a graph with a Graph neural network



E(n) Equivariant Graph Neural Networks; ICLR 21

Three types of equivariance

- Three types of equivariance on a set of particles x:
  - **Translation equivariance**. Translating the input by $g \in R^n$ results in an equivalent translation of the output. Let $x + g$ be shorthand for $(x_1 + g, \cdots, x_M + g)$. Then $y + g = \phi(x + g)$
  - **Rotation (and reflection) equivariance**. For any orthogonal matrix $Q \in R^{n \times n}$, let Qx be shorthand for $(Qx_1, \cdots, Qx_M)$. Then rotating the input results in an equivalent rotation of the output $Qy = (Qx)$.
  - **Permutation equivariance**. Permuting the input results in the same permutation of the output $P(y) = \phi\big(P(x)\big)$, where P is a permutation on the row indexes.

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Equivarient Graph Neural Networks

- Equivariant Graph Convolutional Layer (EGCL) takes as input the set of node embeddings $h^l = \{h_0^l, \cdots, h_{M-1}^l\}$, coordinate embeddings $x^l = \{x_0^l, \cdots, x_{M-1}^l\}$ and edge information $E = (e_{ij})$ and outputs a transformation on h^{l+1} and x^{l+1}. Concisely: h^{l+1}, x^{l+1} = EGCL[h^l, x^l, E].
- The equations that define this layer are the following:

$$\mathbf{m}_{ij} = \phi_e\left(\mathbf{h}_i^l, \mathbf{h}_j^l, \left\|\mathbf{x}_i^l - \mathbf{x}_j^l\right\|^2, a_{ij}\right)$$

$$\mathbf{x}_i^{l+1} = \mathbf{x}_i^l + C\sum_{j\neq i}\left(\mathbf{x}_i^l - \mathbf{x}_j^l\right)\phi_x\left(\mathbf{m}_{ij}\right)$$

$$\mathbf{m}_i = \sum_{j\neq i}\mathbf{m}_{ij}$$

$$\mathbf{h}_i^{l+1} = \phi_h\left(\mathbf{h}_i^l, \mathbf{m}_i\right)$$

Recent advances of GNNs on molecular learning

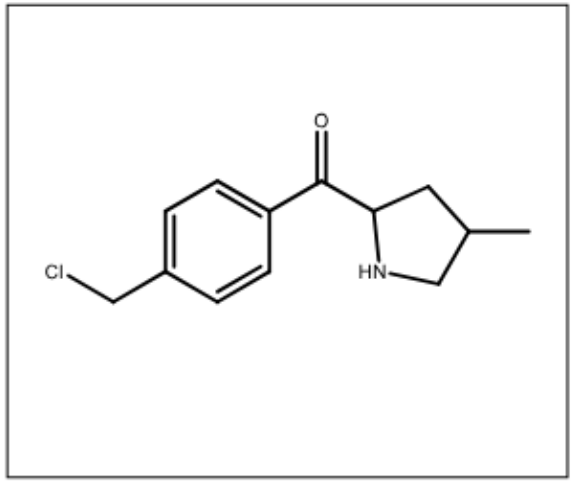
- **Motivation:**
  - Labeled molecules only occupy an extremely small portion of the enormous chemical space since they can only be obtained from wet-lab experiments or quantum chemistry calculations, which are time-consuming and expensive
  - Directly training GNNs on small labelled molecule datasets in a supervised fashion is prone to over-fitting and the trained GNNs can hardly generalize to out-of-distribution data

- **Question**: How to pretrain GNNs on large-scale unlabelled data via self-supervised pretraining?
  - Node-level strategies
  - Contrastive learning strategies
  - Subgraph-level strategies

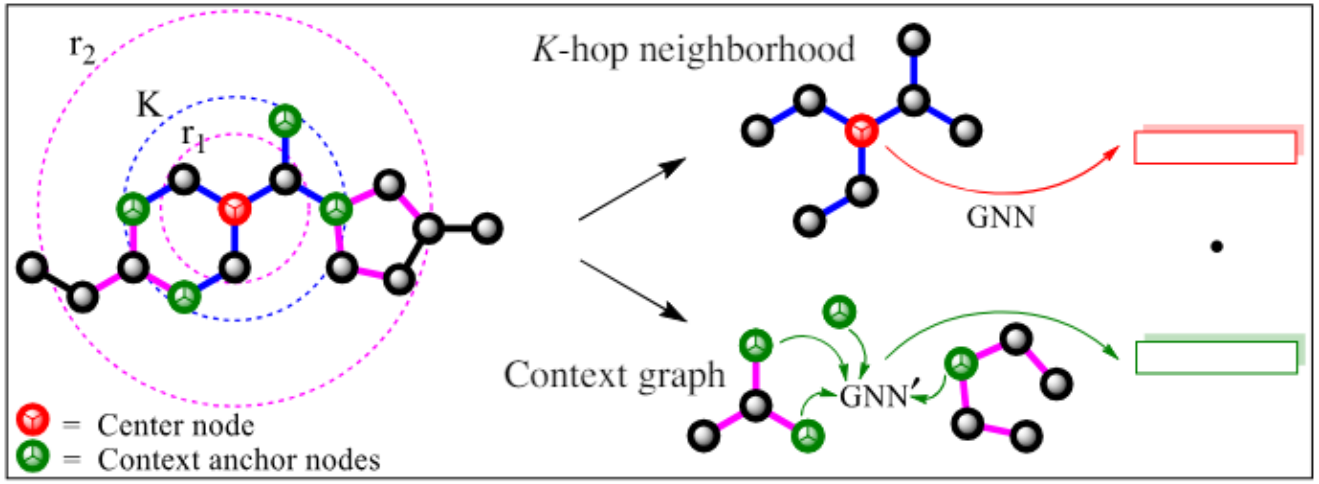
Graph-based strategies for learning molecules

- Node-level strategies: Develop some strategies to relieve the scarce task-specific labels problem
  - In Context Prediction: Nodes in similar contexts obtain nearby embeddings
  - In Attribute Masking: Capture domain knowledge from masking node attributes.

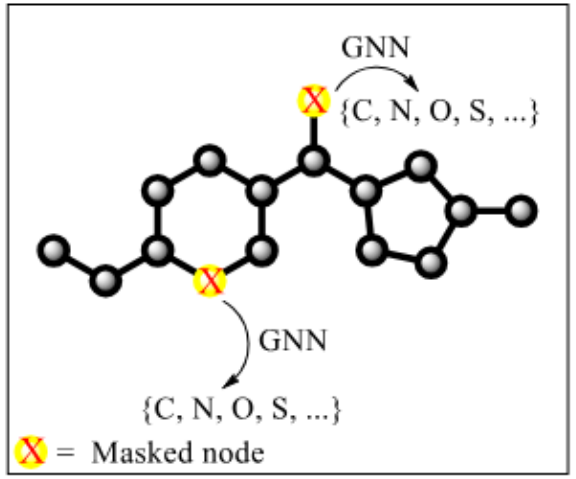
Input molecules



Context Prediction



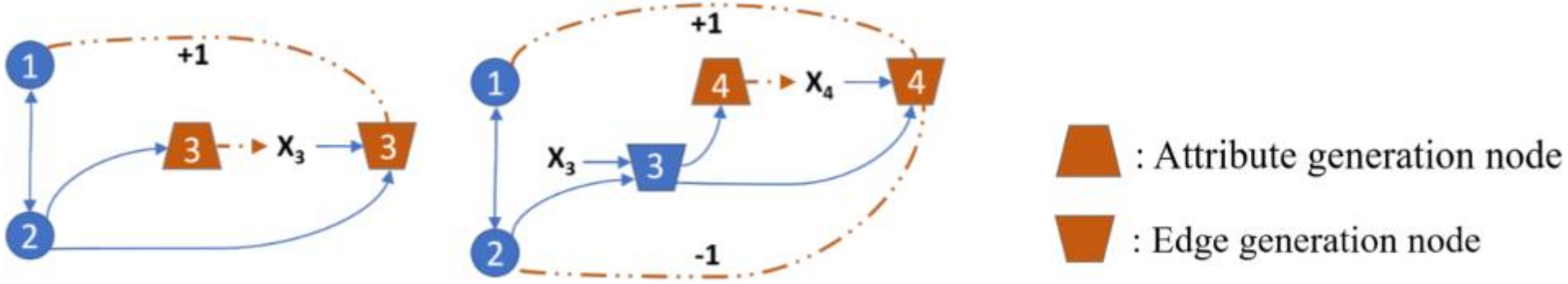
Attribute Masking



Strategies for Pre-training Graph Neural Networks (ICLR 2020)

Graph-based strategies for learning molecules: GPT-GNN

- Pre-train large-scale graph with reconstructing the input graph, which utilizes the unlabelled data for label scarcity issue. Decompose the reconstruction process into two coupled steps:
  - Attribute generation: given observed edges, generate **node attributes**
  - Edge generation: given generated attributes, generate **masked edges**



(d) Generate attribute and masked edges for node 3.

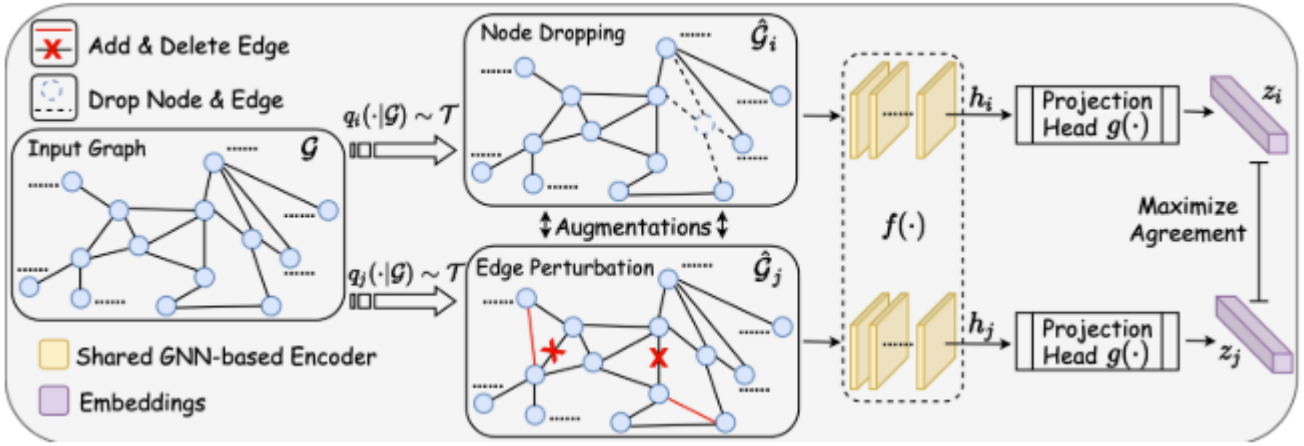
(e) Generate attribute and masked edges for node 4.

Hu Z, Dong Y, Wang K, et al. GPT-GNN: Generative pre-training of graph neural networks, KDD2020

Graph-based strategies for learning molecules: GPT-GNN

- Contrastive learning methods: Design types of graph augmentations to incorporate various impacts in different settings

Data augmentation	Type	Underlying Prior
Node dropping	Nodes, edges	Vertex missing does not alter semantics.
Edge perturbation	Edges	Semantic robustness against connectivity variations.
Attribute masking	Nodes	Semantic robustness against losing partial attributes.
Subgraph	Nodes, edges	Local structure can hint the full semantics.



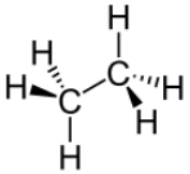
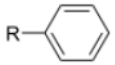
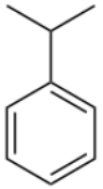
Contrastive Loss:

$$\ell_n = -\log \frac{\exp(\mathrm{sim}(\boldsymbol{z}_{n,i}, \boldsymbol{z}_{n,j})/\tau)}{\sum_{n'=1, n' \neq n}^{N} \exp(\mathrm{sim}(\boldsymbol{z}_{n,i}, \boldsymbol{z}_{n',j})/\tau)}$$

GCC: Graph Contrastive Coding for Graph Neural Network Pre-Training (Qiu et al. KDD2020)
Graph Contrastive Learning with Augmentations (You et al. NIPS2020)
Graph Contrastive Learning with Adaptive Augmentation (Zhu et al. WWW2021)

Graph-based strategies for learning molecules

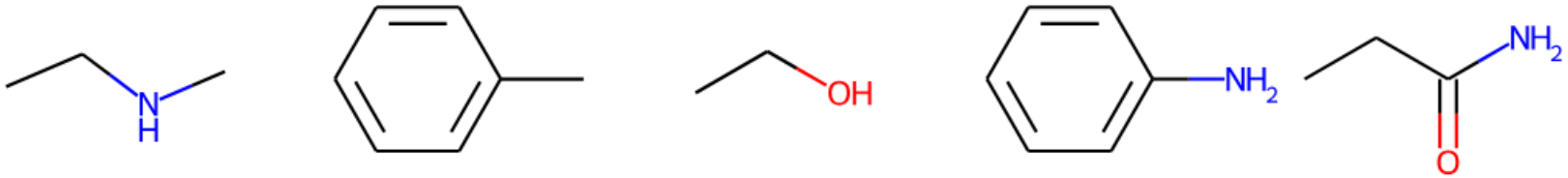
- Limitations of node-level strategies:
  - Models can not be expressive to capture directly the functional groups (characteristic chemical reactions)
  - Contrastive learning strategies can destroy the molecular structures by data augmentation
  - E.g., Hydrocarbons are a class of molecule that is defined by functional groups called hydrocarbyls that contain only carbon and hydrogen

Chemical class	Group	Formula	Structural Formula	Prefix	Suffix	Example
Alkane	Alkyl	$R(CH_2)_nH$		alkyl-	-ane	 Ethane
Alkene	Alkenyl	$R_2C{=}CR_2$		alkenyl-	-ene	Ethylene *(Ethene)*
Alkyne	Alkynyl	RC≡CR'	$R{-}C{\equiv}C{-}R'$	alkynyl-	-yne	$H{-}C{\equiv}C{-}H$ Acetylene *(Ethyne)*
Benzene derivative	Phenyl	RC_6H_5 RPh		phenyl-	- benzene	 Cumene *(Isopropylbenzene)*

Graph-based strategies for learning molecules

- Subgraph-level strategies: Leverage domain knowledge to enhance GNNs for capturing functional groups (subgraphs)
  - capture the regularities of atomic combinations
  - discover repetitive patterns in molecules, thus more capable of generating realistic molecules

- It can reflect chemical properties, since it has been revealed that the chemical properties are closely related to certain subgraphs
  - E.g., The frequent subgraphs are discovered in ZINC dataset:



83.9% 60.5% 41.3% 32.6% 27.5%

Molecule Generation by Principal Subgraph Mining and Assembling, (NeurIPS 2022)

Prior Knowledge to capture functional groups

- Leverage domain knowledge to enhance GNNs for capturing functional groups (subgraphs).
  - capture the regularities of atomic combinations
  - discover repetitive patterns in molecules, thus more capable of generating realistic molecules

- It is able to reflect chemical properties, since it has been revealed that the chemical properties are closely related to certain subgraphs

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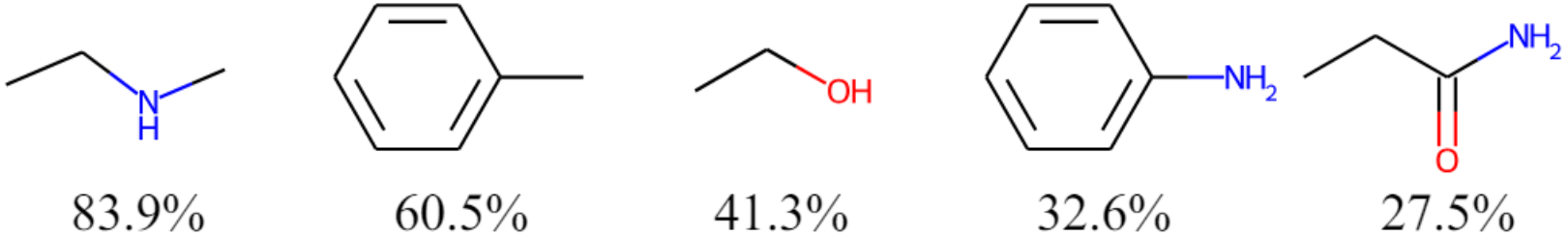
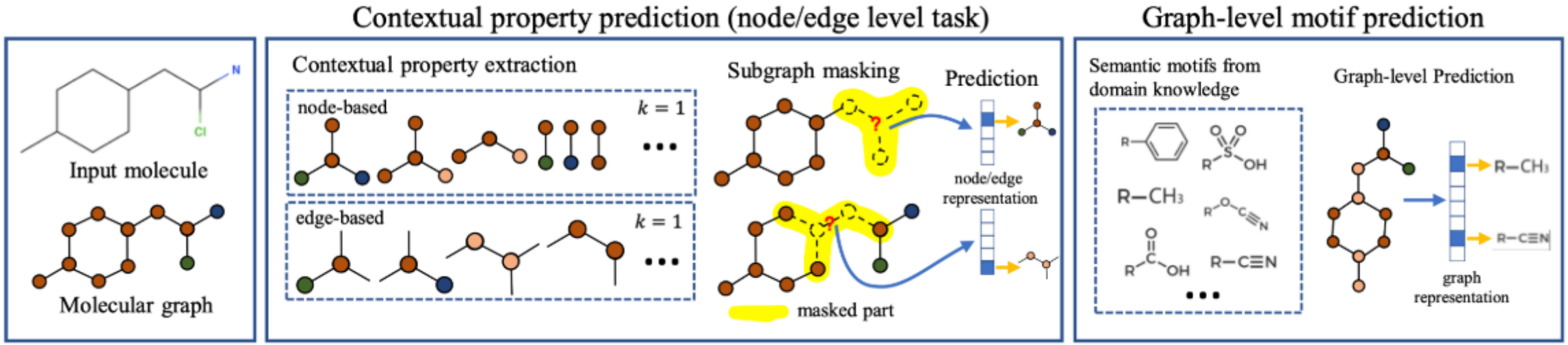


Molecule Generation by Principal Subgraph Mining and Assembling, (NeurIPS 2022)

Graph-based strategies for learning molecules

- GROVER: Node-level and Subgraph-level combination
- Designed self-supervised tasks in node-level, edge-level and graph-level, learn rich **structural** and **semantic** information of molecules
  - Contextual property prediction: predict **masked node/edge attributes** set
  - Motif prediction : predict the classes of **the motif** that occur in each given molecule

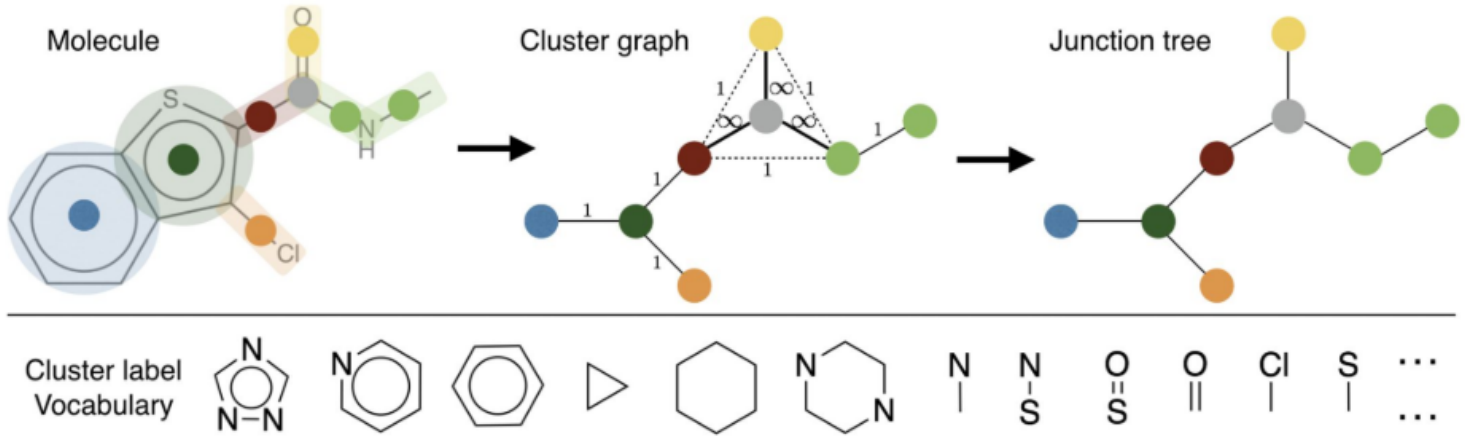


Yu Rong, Yatao Bian, et al. Self-Supervised Graph Transformer on Large-Scale Molecular Data. (NeuIPS2020)

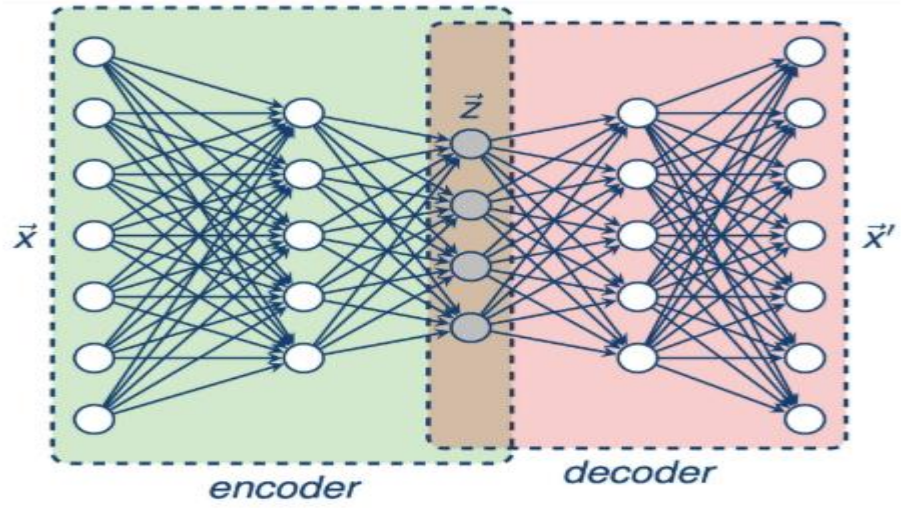
Graph-based strategies for learning molecules

- JTVAE: Clustering atoms to generate tree: Generate molecular graphs iteratively, guided by the molecule's substructures, from a variational representation
- Perform targeted optimization within the representation space

- **Key idea:** Clustering Atoms to generate Junction tree



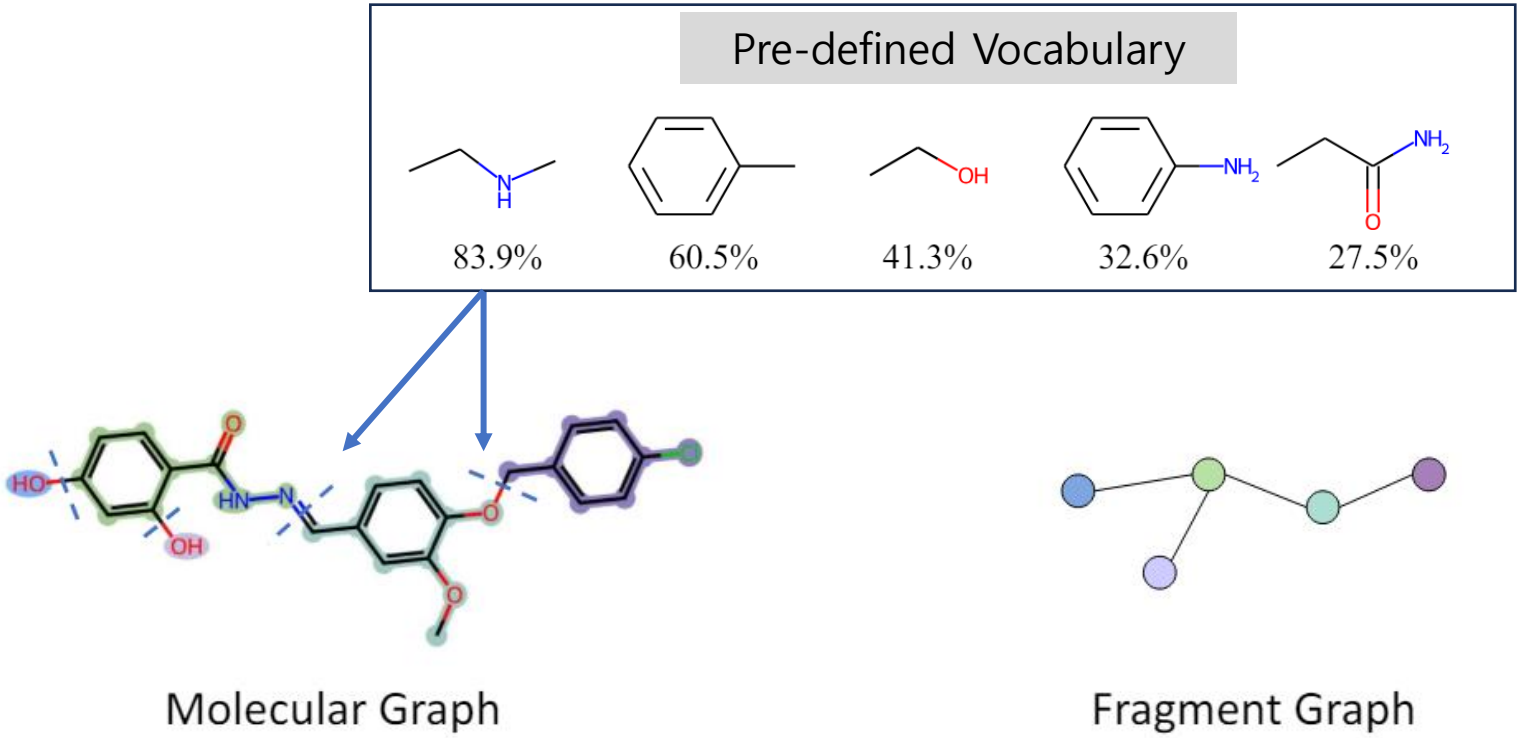
Variational Autoencoder



Junction Tree Variational Autoencoder (Jin et al., ICML 2018)

Graph-based strategies for learning molecules

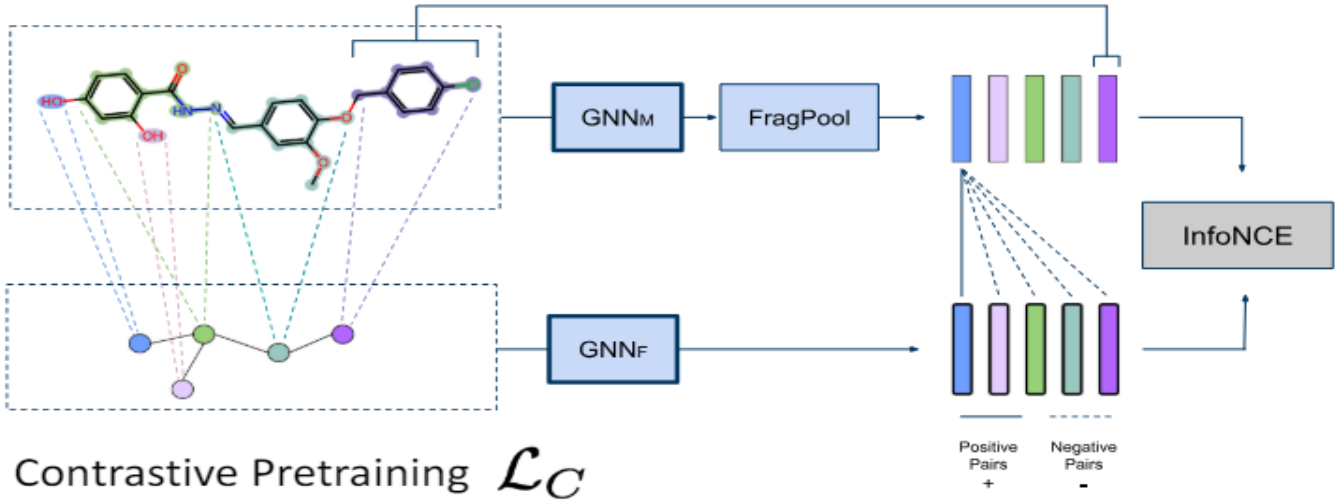
- Fragment-based Pretraining and Finetuning on Molecular Graphs (NeuIPS'23)
- A pretraining task at the subgraph level.
- Prepare molecular graphs and fragment graphs and utilizing them in both pretraining and finetuning. Molecules are fragmented according to a vocabulary extracted via existing frequency-based methods.



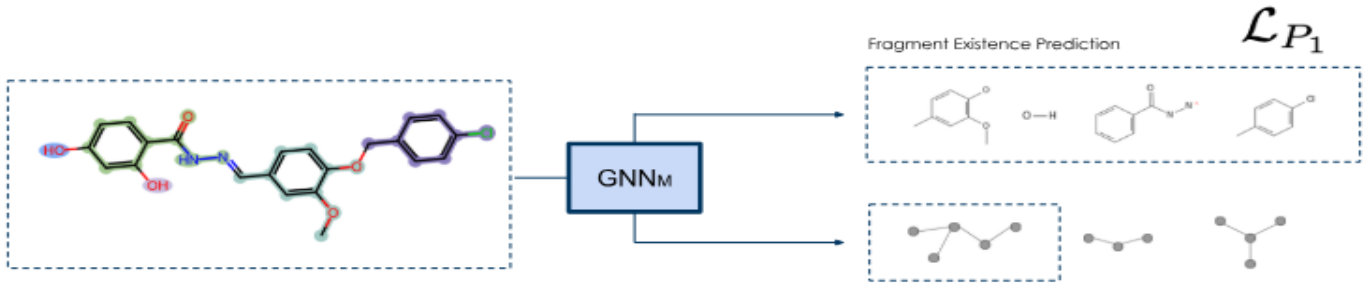
Fragment-based Pretraining and Finetuning on Molecular Graphs, (Kha et al.,NeuIPS'23)

Graph-based strategies for learning molecules

- Fragment-based Pretraining and Finetuning on Molecular Graphs (NeuIPS'23)
- Fragmenting original molecules as a bags of groups:
  - Use Prior Knowledge to build the Dictionary to decompose molecules as a bag of functional groups



Contrastive Pretraining $\mathcal{L}_C$



Predictive Pretraining $\mathcal{L}_P$

GNNM — Process molecular graphs, encoding local patterns

GNNF — Process fragment graphs, encoding global patterns

Contrastively enforces both local and global s tructural patterns into node embeddings

L_{P_1} Fragment existence prediction

L_{P_2} Main structure backbone prediction

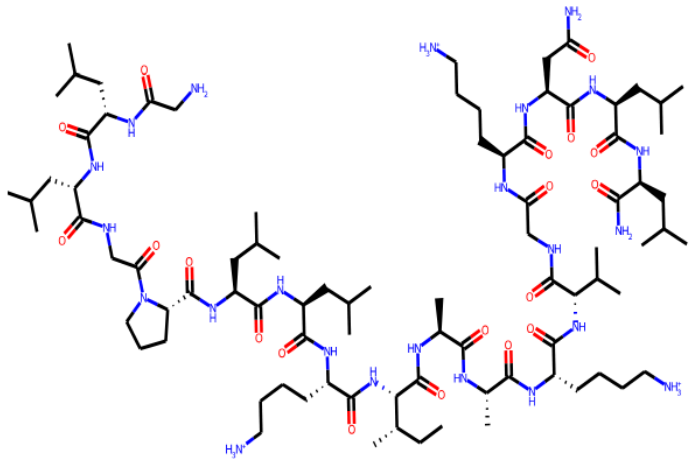
Combined Pretraining

$$\mathcal{L}_P = \mathcal{L}_{P_1} + \mathcal{L}_{P_2}$$

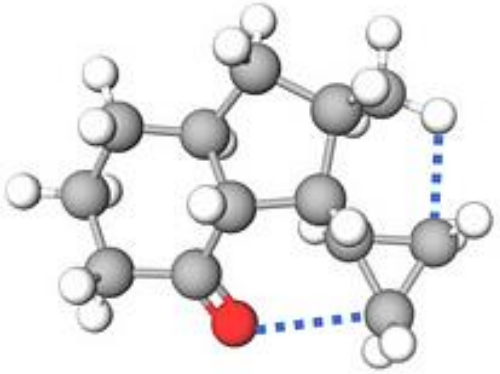
$$\mathcal{L} = \alpha\mathcal{L}_P + (1-\alpha)\mathcal{L}_C$$

Challenging in molecule learning

- Efficiency on large-scale graphs: The molecules in the existing benchmarks are of small sizes, *i.e.*, number of nodes.
  - If graph sizes are large, GNNs would potentially lose out signals flowing from distant nodes due to over-squashing.
  - It is challenging to capture information exchange among distant nodes, where interactions are required far away from the near-local neighbourhood



Long peptide sequence: “GLLGPLLKIAAKVGKNLL”

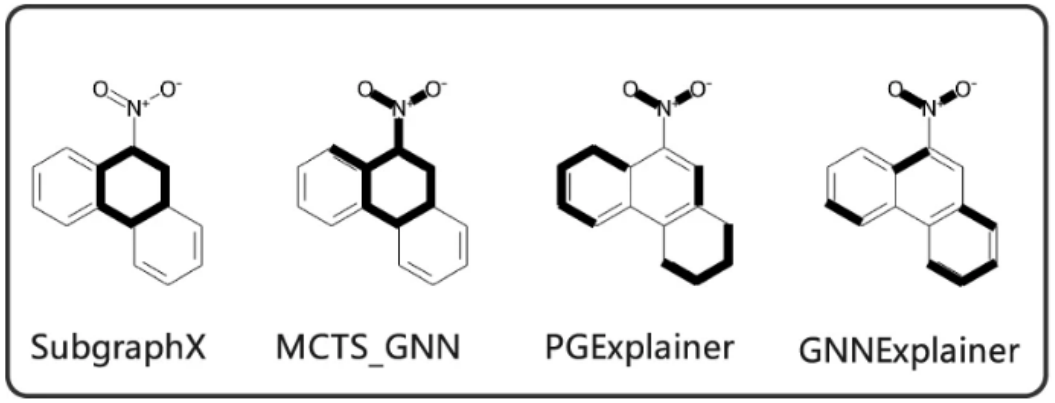


A molecule with long range interactions.
The dotted lines show 3D atomic contact.

Long Range Graph Benchmark, (NeurIPS 2022)

Challenging in molecule learning

- Explainability: Unlike images and texts, molecules are not grid-like data, which means there is no locality information and each node has different numbers of neighbors.
  - The adjacency matrices represent the topology information and only contain discrete values. The discrete adjacency matrices cannot be optimized in the same manner.
  - It is in general difficult to understand the reasoning which leads to the models' output predictions.
    - E.g., Different explainable strategies can provide different solutions

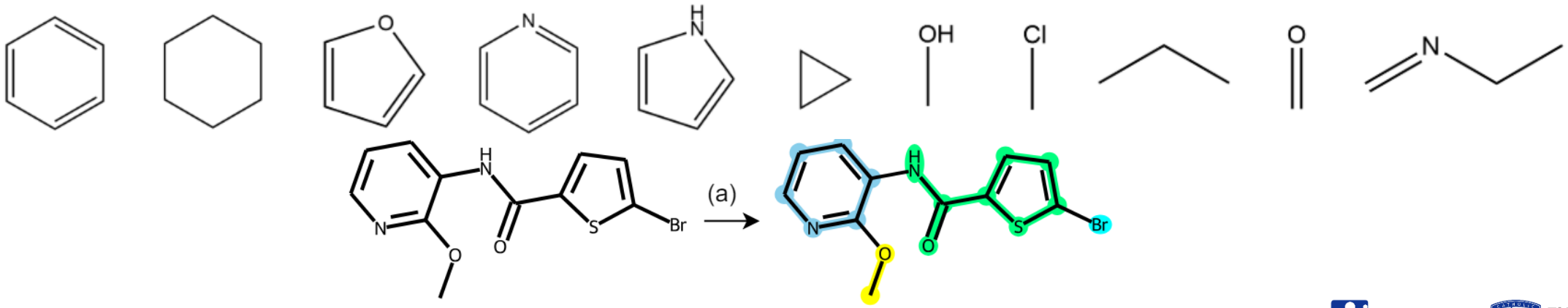


Gnnexplainer: generating explanations for graph neural networks. (NeuIPS 2019)
Parameterized explainer for graph neural network. (*Adv. neural Inf. Process. Syst* 2020)
On explainability of graph neural networks via subgraph explorations. (PMLR 2021)

Challenging in molecule learning

- The use of domain knowledge: To capture functional groups: existing studies focus on extracting frequent subgraphs to generate dictionary
- **It is challenging**:
  - More than one solution exists to break the molecules into bags of fragments.
  - The optimal settings are not easy to find (dictionary size, rules,…). There exist more than one solutions to break molecules
  - It hard to generalize large-scale molecules in varied domains

Motif Vocabulary



Motif-based Graph Self-Supervised Learning for Molecular Property Prediction (NeurIPS 2021)

Q
A